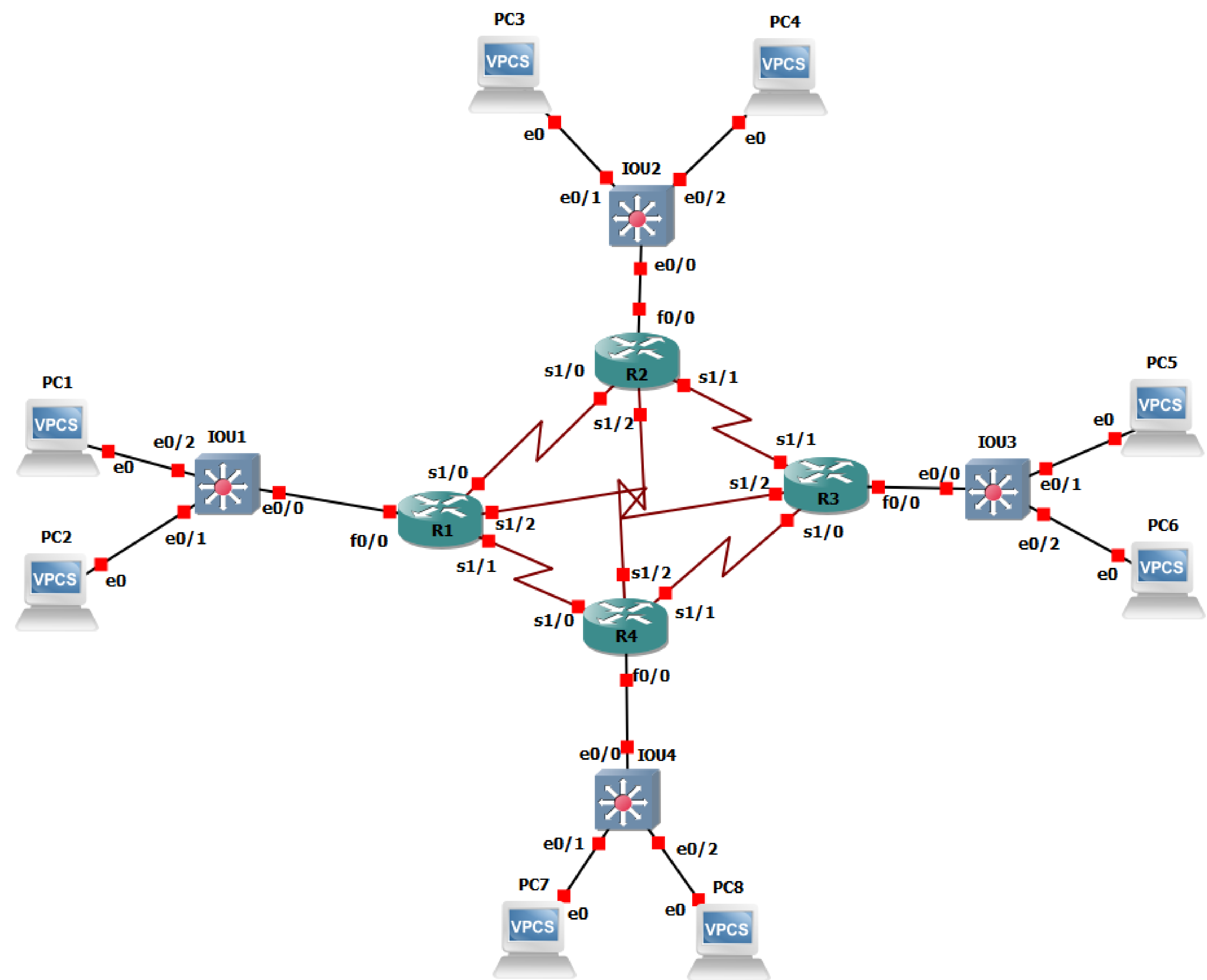




LSA

```

  ▾ LS Update Packet
    Number of LSAs: 1
    ▾ LSA-type 1 (Router-LSA), len 108
      .000 0000 0000 0001 = LS Age (seconds): 1
      0... .... .... .... = Do Not Age Flag: 0
      > Options: 0x22, (DC) Demand Circuits, (E) External Routing
      LS Type: Router-LSA (1)
      <Router LSA: True>
      Link State ID: 166.15.112.1
      Advertising Router: 166.15.112.1
      Sequence Number: 0x80000003
      Checksum: 0x96ce
      Length: 108
      > Flags: 0x00
      Number of Links: 7
      > Type: PTP      ID: 166.15.160.2   Data: 166.15.112.1   Metric: 2
      > Type: Stub     ID: 166.15.112.0   Data: 255.255.240.0 Metric: 2
      > Type: PTP      ID: 166.15.160.1   Data: 166.15.96.1   Metric: 2
      > Type: Stub     ID: 166.15.96.0    Data: 255.255.240.0 Metric: 2
      > Type: PTP      ID: 166.15.144.1   Data: 166.15.80.1   Metric: 1
      > Type: Stub     ID: 166.15.80.0    Data: 255.255.240.0 Metric: 1
      > Type: Stub     ID: 166.15.32.0    Data: 255.255.240.0 Metric: 1

```

R1 Routing Table

Network	Next Hop	Metric	Interface

166.15.160.0	166.15.112.2	110/5	Serial1/1
	166.15.96.2	110/5	Serial1/2
166.15.128.0	166.15.80.2	110/6	Serial1/0
166.15.144.0	166.15.80.2	110/8	Serial1/0
166.15.32.0	Directly connected		FastEthernet0/0
166.15.48.0	166.15.112.2	110/3	Serial1/1
166.15.16.0	166.15.96.2	110/3	Serial1/2
166.15.96.0	Directly connected		Serial1/2
166.15.112.0	Directly connected		Serial1/1
166.15.64.0	166.15.80.2	110/2	Serial1/0
166.15.80.0	Directly connected		Serial1/0

Router 1

```
Config t
Int f0/0 ### N1
Ip address 166.15.32.1 255.255.240.0
no shutdown
int s1/0 ## to R2
ip address 166.15.80.1 255.255.240.0
ip ospf cost 1
no shutdown
int s1/2 ## to R3
ip address 166.15.96.1 255.255.240.0
ip ospf cost 2
no shutdown
int s1/1 ## to R4
ip address 166.15.112.1 255.255.240.0
ip ospf cost 2
no shutdown
exit
router ospf 10
network 166.15.32.0 0.0.15.255 area 0
network 166.15.80.0 0.0.15.255 area 0
network 166.15.96.0 0.0.15.255 area 0
network 166.15.112.0 0.0.15.255 area 0
end
copy running-config startup-config
```


LSA

```

  ▾ LS Update Packet
    Number of LSAs: 1
    ▾ LSA-type 1 (Router-LSA), len 108
      .000 0000 0000 0001 = LS Age (seconds): 1
      0... .... .... .... = Do Not Age Flag: 0
      > Options: 0x22, (DC) Demand Circuits, (E) External Routing
      LS Type: Router-LSA (1)
      <Router LSA: True>
      Link State ID: 166.15.144.1
      Advertising Router: 166.15.144.1
      Sequence Number: 0x80000003
      Checksum: 0x365d
      Length: 108
      > Flags: 0x00
      Number of Links: 7
      > Type: PTP      ID: 166.15.160.2   Data: 166.15.144.1   Metric: 7
      > Type: Stub     ID: 166.15.144.0   Data: 255.255.240.0 Metric: 7
      > Type: PTP      ID: 166.15.160.1   Data: 166.15.128.1   Metric: 5
      > Type: Stub     ID: 166.15.128.0   Data: 255.255.240.0 Metric: 5
      > Type: PTP      ID: 166.15.112.1   Data: 166.15.80.2    Metric: 1
      > Type: Stub     ID: 166.15.80.0    Data: 255.255.240.0 Metric: 1
      > Type: Stub     ID: 166.15.64.0    Data: 255.255.240.0 Metric: 1

```

R2 Routing Table

Network	Next Hop	Metric	Interface

166.15.160.0	166.15.80.1	110/6	Serial1/0
166.15.128.0	Directly connected		Serial1/1
166.15.144.0	Directly connected		Serial1/2
166.15.32.0	166.15.80.1	110/2	Serial1/0
166.15.48.0	166.15.80.1	110/4	Serial1/0
166.15.16.0	166.15.80.1	110/4	Serial1/0
166.15.96.0	166.15.80.1	110/3	Serial1/0
166.15.112.0	166.15.80.1	110/3	Serial1/0
166.15.64.0	Directly connected		FastEthernet0/0
166.15.80.0	Directly connected		Serial1/0

Router 2

```
config t
int f0/0 ### N2
ip address 166.15.64.1 255.255.240.0
no shutdown
int s1/0 ## to R1
ip address 166.15.80.2 255.255.240.0
ip ospf cost 1
no shutdown
int s1/1 ## to R3
ip address 166.15.128.1 255.255.240.0
ip ospf cost 5
no shutdown
int s1/2 ## to R4
ip address 166.15.144.1 255.255.240.0
ip ospf cost 7
no shutdown
exit
router ospf 10
network 166.15.64.0 0.0.15.255 area 0
network 166.15.80.0 0.0.15.255 area 0
network 166.15.128.0 0.0.15.255 area 0
network 166.15.144.0 0.0.15.255 area 0
end
copy running-config startup-config
```


LSA

```

  ▾ LS Update Packet
    Number of LSAs: 1
    ▾ LSA-type 1 (Router-LSA), len 108
      .000 0000 0000 0010 = LS Age (seconds): 2
      0... .... .... .... = Do Not Age Flag: 0
      > Options: 0x22, (DC) Demand Circuits, (E) External Routing
      LS Type: Router-LSA (1)
      <Router LSA: True>
      Link State ID: 166.15.160.1
      Advertising Router: 166.15.160.1
      Sequence Number: 0x80000003
      Checksum: 0x0474
      Length: 108
      > Flags: 0x00
      Number of Links: 7
      > Type: PTP      ID: 166.15.160.2   Data: 166.15.160.1   Metric: 3
      > Type: Stub     ID: 166.15.160.0   Data: 255.255.240.0 Metric: 3
      > Type: PTP      ID: 166.15.144.1   Data: 166.15.128.2   Metric: 5
      > Type: Stub     ID: 166.15.128.0   Data: 255.255.240.0 Metric: 5
      > Type: PTP      ID: 166.15.112.1   Data: 166.15.96.2    Metric: 2
      > Type: Stub     ID: 166.15.96.0    Data: 255.255.240.0 Metric: 2
      > Type: Stub     ID: 166.15.16.0    Data: 255.255.240.0 Metric: 1

```

R3 Routing Table

Network	Next Hop	Metric	Interface

166.15.160.0	Directly connected		Serial1/0
166.15.128.0	Directly connected		Serial1/1
166.15.144.0	166.15.160.2	110/10	Serial1/0
	166.15.96.1	110/10	Serial1/2
166.15.32.0	166.15.96.1	110/3	Serial1/2
166.15.48.0	166.15.160.2	110/4	Serial1/0
166.15.16.0	Directly connected		FastEthernet0/0
166.15.96.0	Directly connected		Serial1/2
166.15.112.0	166.15.96.1	110/4	Serial1/2
166.15.64.0	166.15.96.1	110/4	Serial1/2
166.15.80.0	166.15.96.1	110/3	Serial1/2

Router 3

```
config t
int f0/0 ### N3
ip address 166.15.16.1 255.255.240.0
no shutdown
int s1/2 ## to R1
ip address 166.15.96.2 255.255.240.0
ip ospf cost 2
no shutdown
int s1/1 ## to R2
ip address 166.15.128.2 255.255.240.0
ip ospf cost 5
no shutdown
int s1/0 ## to R4
ip address 166.15.160.1 255.255.240.0
ip ospf cost 3
no shutdown
exit
router ospf 10
network 166.15.16.0 0.0.15.255 area 0
network 166.15.96.0 0.0.15.255 area 0
network 166.15.128.0 0.0.15.255 area 0
network 166.15.160.0 0.0.15.255 area 0
end
copy running-config startup-config
```


LSA

```

  ▾ LS Update Packet
    Number of LSAs: 1
    ▾ LSA-type 1 (Router-LSA), len 108
      .000 0000 0000 0010 = LS Age (seconds): 2
      0... .... .... .... = Do Not Age Flag: 0
      > Options: 0x22, (DC) Demand Circuits, (E) External Routing
      LS Type: Router-LSA (1)
      <Router LSA: True>
      Link State ID: 166.15.160.2
      Advertising Router: 166.15.160.2
      Sequence Number: 0x80000003
      Checksum: 0x30e1
      Length: 108
      > Flags: 0x00
      Number of Links: 7
      > Type: PTP      ID: 166.15.160.1   Data: 166.15.160.2   Metric: 3
      > Type: Stub     ID: 166.15.160.0   Data: 255.255.240.0 Metric: 3
      > Type: PTP      ID: 166.15.144.1   Data: 166.15.144.2   Metric: 7
      > Type: Stub     ID: 166.15.144.0   Data: 255.255.240.0 Metric: 7
      > Type: PTP      ID: 166.15.112.1   Data: 166.15.112.2   Metric: 2
      > Type: Stub     ID: 166.15.112.0   Data: 255.255.240.0 Metric: 2
      > Type: Stub     ID: 166.15.48.0    Data: 255.255.240.0 Metric: 1

```

R4 Routing Table

Network	Next Hop	Metric	Interface

166.15.160.0	Directly connected		Serial1/1
166.15.128.0	166.15.160.1	110/8	Serial1/1
	166.15.112.1	110/8	Serial1/0
166.15.144.0	Directly connected		Serial1/2
166.15.32.0	166.15.112.1	110/3	Serial1/0
166.15.48.0	Directly connected		FastEthernet0/0
166.15.16.0	166.15.160.1	110/4	Serial1/1
166.15.96.0	166.15.112.1	110/4	Serial1/0
166.15.112.0	Directly connected		Serial1/0
166.15.64.0	166.15.112.1	110/4	Serial1/0
166.15.80.0	166.15.112.1	110/3	Serial1/0

Router 4

```
config t
int f0/0 ### N4
ip address 166.15.48.1 255.255.240.0
no shutdown
int s1/0 ## to R1
ip address 166.15.112.2 255.255.240.0
ip ospf cost 2
no shutdown
int s1/2 ## to R2
ip address 166.15.144.2 255.255.240.0
ip ospf cost 7
no shutdown
int s1/1 ## to R3
ip address 166.15.160.2 255.255.240.0
ip ospf cost 3
no shutdown
exit
router ospf 10
network 166.15.48.0 0.0.15.255 area 0
network 166.15.112.0 0.0.15.255 area 0
network 166.15.144.0 0.0.15.255 area 0
network 166.15.166.0 0.0.15.255 area 0
end
copy running-config startup-config
```